

Table 7-25. Get VCS Domain Validation SV State Request Payload

| Byte Offset | Length in Bytes | Description                           |
|-------------|-----------------|---------------------------------------|
| 0h          | 1               | <b>VCS ID:</b> Index of VCS to query. |

Table 7-26. Get VCS Domain Validation SV State Response Payload

| Byte Offset | Length in Bytes | Description                                                                                                                                                                        |
|-------------|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0h          | 1               | <b>Secret Value State:</b> State of the secret value: <ul style="list-style-type: none"> <li>00h = Not set</li> <li>01h = Set</li> <li>All other encodings are reserved</li> </ul> |

#### 7.6.7.1.8 Get Domain Validation SV (Opcode 5107h)

This command is used by the FM to retrieve the secret value from a VCS.

Possible Command Return Codes:

- Success
- Internal Error
- Retry Required

Command Effects:

- None

Table 7-27. Get Domain Validation SV Request Payload

| Byte Offset | Length in Bytes | Description                           |
|-------------|-----------------|---------------------------------------|
| 0h          | 1               | <b>VCS ID:</b> Index of VCS to query. |

Table 7-28. Get Domain Validation SV Response Payload

| Byte Offset | Length in Bytes | Description                                                           |
|-------------|-----------------|-----------------------------------------------------------------------|
| 0h          | 10h             | <b>Secret Value:</b> UUID used to uniquely identify a host hierarchy. |

#### 7.6.7.2 Virtual Switch Command Set

This command set is supported only by the CXL switch. It is required for switches that support more than one VCS. The following commands are defined:

Table 7-29. Virtual Switch Command Set Requirements

| Command Name                | Requirement <sup>1</sup> |
|-----------------------------|--------------------------|
| Get Virtual CXL Switch Info | M                        |
| Bind vPPB                   | O                        |
| Unbind vPPB                 | O                        |
| Generate AER Event          | O                        |

1. M = Mandatory, O = Optional.

#### 7.6.7.2.1 Get Virtual CXL Switch Info (Opcode 5200h)

This command retrieves information on a specified number of VCSs in the switch. Because of the possibility of variable numbers of vPPBs within each VCS, the returned array has variably sized elements.

Possible Command Return Codes:

- Success
- Invalid Input
- Internal Error
- Retry Required
- Invalid Payload Length

Command Effects:

- None

**Table 7-30. Get Virtual CXL Switch Info Request Payload**

| Byte Offset | Length in Bytes | Description                                                                                                                                                                                                                                                                        |
|-------------|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0h          | 1               | <b>Start vPPB:</b> Specifies the ID of the first vPPB for each VCS to include in the vPPB information list in the response (bytes 4 – 7 in <a href="#">Table 7-32</a> ). This enables compatibility with devices that have small maximum command message sizes.                    |
| 1h          | 1               | <b>vPPB List Limit:</b> The maximum number of vPPB information entries to include in the response (bytes 4 – 7 in <a href="#">Table 7-32</a> ). This enables compatibility with devices that have small maximum command message sizes. This field shall have a minimum value of 1. |
| 2h          | 1               | <b>Number of VCSs:</b> Number of VCSs requested. This field shall have a minimum value of 1.                                                                                                                                                                                       |
| 3h          | Number of VCSs  | <b>VCS ID List:</b> 1-byte ID of requested VCS, repeated <b>Number of VCSs</b> times.                                                                                                                                                                                              |

**Table 7-31. Get Virtual CXL Switch Info Response Payload**

| Byte Offset | Length in Bytes | Description                                                                                                                         |
|-------------|-----------------|-------------------------------------------------------------------------------------------------------------------------------------|
| 0h          | 1               | <b>Number of VCSs:</b> Number of VCS information blocks returned.                                                                   |
| 1h          | 3               | <b>Reserved</b>                                                                                                                     |
| 4h          | Varies          | <b>VCS Information List:</b> VCS information block as defined in <a href="#">Table 7-32</a> , repeated <b>Number of VCSs</b> times. |

Table 7-32. Get Virtual CXL Switch Info VCS Information Block Format

| Byte Offset                         | Length in Bytes | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|-------------------------------------|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0h                                  | 1               | <b>Virtual CXL Switch ID</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 1h                                  | 1               | <b>VCS State:</b> Current state of the VCS: <ul style="list-style-type: none"> <li>• 00h = Disabled</li> <li>• 01h = Enabled</li> <li>• FFh = Invalid VCS ID; all subsequent field values are invalid</li> <li>• All other encodings are reserved</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 2h                                  | 1               | <b>USP ID:</b> Physical port ID of the current VCS's Upstream Port, or the current VCS's fabric physical port ID of a Downstream ES VCS. Valid only when the VCS is enabled.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 3h                                  | 1               | <b>Number of vPPBs:</b> Total number of vPPBs in the VCS. This value may be larger than the vPPB List Limit field specified in the request. In this case, the length of vPPB information list, starting at byte 4, is defined by "vPPB List Limit", not by this field. vPPB information list consists of vPPB List Entry Count number of entries and each entry is 4B in length. vPPB List Entry Count = min(vPPB List Limit, Number of vPPBs).                                                                                                                                                                                                                                                                                                                           |
| 4h                                  | 1               | <b>vPPB[Start vPPB] Binding Status</b> <ul style="list-style-type: none"> <li>• 00h = Unbound</li> <li>• 01h = Bind or unbind in progress</li> <li>• 02h = Bound Physical Port</li> <li>• 03h = Bound LD</li> <li>• 04h = Bound PID</li> <li>• All other encodings are reserved</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 5h                                  | 2               | For PBR Switches when Binding Status is 02h or 03h and for HBR Switches: <ul style="list-style-type: none"> <li>• Bits[7:0]: <b>vPPB[Start vPPB] Bound Port ID:</b> Physical port number of the bound port. Valid only when Binding Status is 02h or 03h.</li> <li>• Bits[15:8]: <b>vPPB[Start vPPB] Bound LD ID:</b> ID of the LD that is bound to the port from the MLD on an associated physical port. Valid only when vPPB[Start vPPB] Binding Status is 03h; otherwise, the value is FFh.</li> </ul> For PBR Switches when Binding Status is 04h: <ul style="list-style-type: none"> <li>• Bits[11:0]: <b>vPPB[Start vPPB] Bound PID:</b> PID of the bound vPPB, as defined in <a href="#">Section 7.7.12.3</a>.</li> <li>• Bits[15:12]: <b>Reserved.</b></li> </ul> |
| 7h                                  | 1               | <b>Reserved</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| ...                                 | ...             | ...                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 4 + (vPPB List Entry Count - 1) * 4 | 1               | <b>vPPB[Start vPPB + vPPB List Entry Count<sup>1</sup> - 1] Binding Status:</b> As defined above.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 5 + (vPPB List Entry Count - 1) * 4 | 1               | <b>vPPB[Start vPPB + vPPB List Entry Count<sup>1</sup> - 1] Bound Port ID:</b> As defined above.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 6 + (vPPB List Entry Count - 1) * 4 | 1               | <b>vPPB[Start vPPB + vPPB List Entry Count<sup>1</sup> - 1] Bound LD ID:</b> As defined above.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 7 + (vPPB List Entry Count - 1) * 4 | 1               | <b>Reserved</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |

1. The vPPB information list length is defined by the lesser of the vPPB List Limit field in the command request and the Number of vPPBs field in the command response.

#### 7.6.7.2.2 Bind vPPB (Opcode 5201h)

This command performs a binding operation on the specified vPPB. If the bind target is a physical port connected to a Type 1, Type 2, Type 3, or PCIe device or a physical port whose link is down, the specified physical port of the CXL switch is fully bound to the vPPB. If the bind target is a physical port connected to an MLD, then a corresponding LD-ID must also be specified.

All binding operations are executed as background commands. The switch notifies the FM of binding completion through the generation of event records, as defined in [Section 7.6.6](#).

Possible Command Return Codes:

- Background Command Started
- Unsupported
- Invalid Input
- Internal Error
- Retry Required
- Busy

Command Effects:

- Background Operation

**Table 7-33. Bind vPPB Request Payload**

| Byte Offset | Length in Bytes | Description                                                                              |
|-------------|-----------------|------------------------------------------------------------------------------------------|
| 0h          | 1               | <b>Virtual CXL Switch ID</b>                                                             |
| 1h          | 1               | <b>vPPB ID:</b> Index of the vPPB within the VCS specified in the VCS ID.                |
| 2h          | 1               | <b>Physical Port ID</b>                                                                  |
| 3h          | 1               | <b>Reserved</b>                                                                          |
| 4h          | 2               | <b>LD ID:</b> LD-ID if the target port is an MLD port. Must be FFFFh for other EP types. |

#### 7.6.7.2.3 Unbind vPPB (Opcode 5202h)

This command unbinds the physical port or LD from the virtual hierarchy vPPB. All unbinding operations are executed as background commands. The switch notifies the FM of unbinding completion through the generation of event records, as defined in [Section 7.6.6](#).

Possible Command Return Codes:

- Unsupported
- Background Command Started
- Invalid Input
- Internal Error
- Retry Required
- Busy

Command Effects:

- Background Operation

Table 7-34. Unbind vPPB Request Payload

| Byte Offset | Length in Bytes | Description                                                                                                                                                                                                                                                                                                                                            |
|-------------|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0h          | 1               | <b>Virtual CXL Switch ID</b>                                                                                                                                                                                                                                                                                                                           |
| 1h          | 1               | <b>vPPB ID:</b> Index of the vPPB within the VCS specified in the VCS ID.                                                                                                                                                                                                                                                                              |
| 2h          | 1               | <ul style="list-style-type: none"> <li>Bits[3:0]: <b>Unbind Option:</b> <ul style="list-style-type: none"> <li>0h = Wait for port Link Down before unbinding</li> <li>1h = Simulate Managed Hot-Remove</li> <li>2h = Simulate Surprise Hot-Remove</li> <li>All other encodings are reserved</li> </ul> </li> <li>Bits[7:4]: <b>Reserved</b></li> </ul> |

#### 7.6.7.2.4 Generate AER Event (Opcode 5203h)

This command generates an AER event on a specified VCS's vPPB (US vPPB or DS vPPB). The switch must respect the Host's AER mask settings.

Possible Command Return Codes:

- Success
- Unsupported
- Invalid Input
- Internal Error
- Retry Required

Command Effects:

- None

Table 7-35. Generate AER Event Request Payload

| Byte Offset | Length in Bytes | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|-------------|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0h          | 1               | <b>Virtual CXL Switch ID</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 1h          | 1               | <b>vPPB Instance:</b> The value of 0 represents USP. The values of 1 and above represent the DSP vPPBs in increasing Device Number, Function Number order, as defined in Section 7.1.4.                                                                                                                                                                                                                                                                                                                                                                                                          |
| 2h          | 2               | <b>Reserved</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 4h          | 4               | <b>AER Error</b> <ul style="list-style-type: none"> <li>Bits[4:0]: <ul style="list-style-type: none"> <li>If Severity=0, bit position of the error type in the AER Correctable Error Status register, as defined in PCIe Base Specification</li> <li>If Severity=1, bit position of the error type in the AER Uncorrectable Error Status register, as defined in PCIe Base Specification</li> </ul> </li> <li>Bits[30:5]: <b>Reserved</b></li> <li>Bit[31]: <b>Severity</b> <ul style="list-style-type: none"> <li>0 = Correctable Error</li> <li>1 = Uncorrectable Error</li> </ul> </li> </ul> |
| 8h          | 20h             | <b>AER Header:</b> TLP Header to place in AER registers, as defined in PCIe Base Specification.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

#### 7.6.7.3 MLD Port Command Set

This command set is applicable to CXL switches and MLDs. The following commands are defined:

Table 7-36. MLD Port Command Set Requirements

| Command Name                         | Requirement           |                   |
|--------------------------------------|-----------------------|-------------------|
|                                      | Switches <sup>1</sup> | MLDs <sup>1</sup> |
| Tunnel Management Command            | M                     | O                 |
| Send LD CXL.io Configuration Request | M                     | P                 |
| Send LD CXL.io Memory Request        | M                     | P                 |

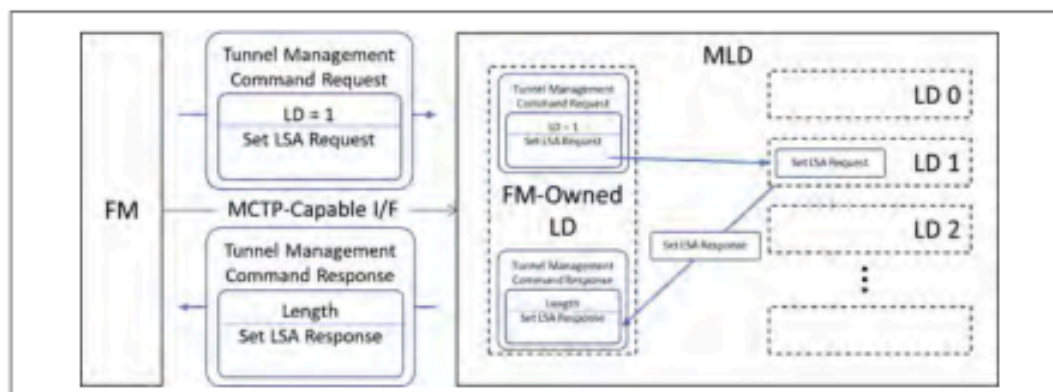
1. M = Mandatory, O = Optional, P = Prohibited.

### 7.6.7.3.1 Tunnel Management Command (Opcode 5300h)

This command tunnels the provided command to LD FFFFh of the MLD on the specified port, using the transport defined in Section 7.6.3.1.

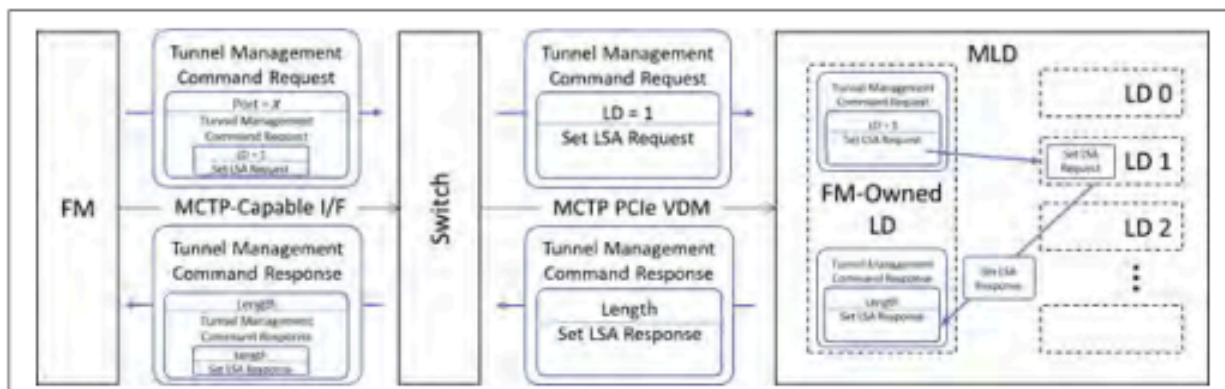
When sent to an MLD, this provided command is tunneled by the FM-owned LD to the specified LD, as illustrated in Figure 7-22 of a “Set LSA Request” being tunneled to LD 1 in an MLD.

Figure 7-22. Tunneling Commands to an LD in an MLD



The Management Command input payload field includes the tunneled command encapsulated in the CCI Message Format, as defined in Figure 7-19. This can include an additional layer of tunneling for commands issued to LDs in an MLD that is accessible only through a CXL switch's MLD Port, as illustrated in Figure 7-23.

Figure 7-23. Tunneling Commands to an LD in an MLD through a CXL Switch

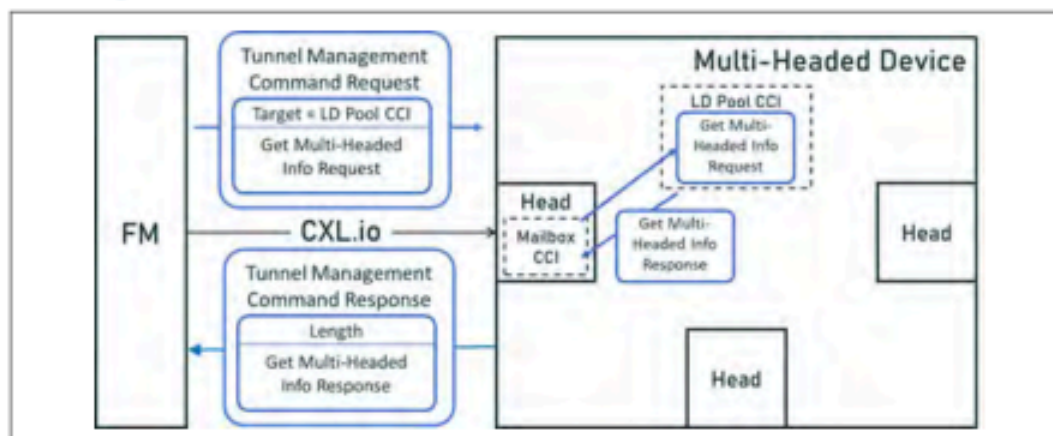




Response size varies, based on the tunneled FM command's definition. Valid targets for the tunneled commands include switch MLD Ports, valid LDs within an MLD, and the LD Pool CCI in a Multi-Headed device. Tunneled commands sent to any other targets shall be discarded and this command shall return an "Invalid Input" return code. The FM-owned LD (LD=FFFFh) is an invalid target in MLDs.

The LD Pool CCI in Multi-Headed devices is targeted using the "Target Type" field, as illustrated in Figure 7-24. This command shall return an "Invalid Input" return code failure if tunneling to the LD Pool CCI is not permitted on the CCI that receives the request.

**Figure 7-24. Tunneling Commands to the LD Pool CCI in a Multi-Headed Device**



A Multi-Headed device shall terminate the processing of a request that includes more than 3 layers of tunneling and return the Unsupported return code.

The Tunnel Management Command itself does not cause any Command Effects, but the Management Command provided in the request will cause Command Effects as per its definition.

Possible Command Return Codes:

- Success
- Unsupported
- Invalid Input
- Internal Error
- Retry Required

Command Effects:

- None

Table 7-37. Tunnel Management Command Request Payload

| Byte Offset | Length in Bytes | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|-------------|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0h          | 1               | <b>Port or LD ID:</b> Egress port ID for commands sent to a switch, or LD-ID for commands sent to an MLD. Valid only when Target Type is 0.                                                                                                                                                                                                                                                                                                                                              |
| 1h          | 1               | <ul style="list-style-type: none"> <li>Bits[3:0]: <b>Target Type:</b> Specifies the type of tunneling target for this command: <ul style="list-style-type: none"> <li>0h = Port or LD based. Indicates that the "Port or LD ID" field is used to determine the target.</li> <li>1h = LD Pool CCI. Indicates that the tunneling target is the LD Pool CCI of a Multi-Headed device.</li> <li>All other encodings are reserved.</li> </ul> </li> <li>Bits[7:4]: <b>Reserved</b></li> </ul> |
| 2h          | 2               | <b>Command Size:</b> Number of valid bytes in <b>Management Command</b> .                                                                                                                                                                                                                                                                                                                                                                                                                |
| 4h          | Varies          | <b>Management Command:</b> Request message formatted in the CCI Message Format as defined in Figure 7-19.                                                                                                                                                                                                                                                                                                                                                                                |

Table 7-38. Tunnel Management Command Response Payload

| Byte offset | Length in Bytes | Description                                                                                              |
|-------------|-----------------|----------------------------------------------------------------------------------------------------------|
| 0h          | 2               | <b>Response Length:</b> Number of valid bytes in <b>Response Message</b> .                               |
| 2h          | 2               | <b>Reserved</b>                                                                                          |
| 4h          | Varies          | <b>Response Message:</b> Response message formatted in the CCI Message Format as defined in Figure 7-19. |

#### 7.6.7.3.2 Send LD CXL.io Configuration Request (Opcode 5301h)

This command allows the FM to read or write the CXL.io Configuration Space of an unbound LD or FMLD. The switch will convert the request into CfgRd/CfgWr TLPs to the target device. Invalid Input Return Code shall be generated if the requested LD is bound.

Possible Command Return Codes:

- Success
- Unsupported
- Invalid Input
- Internal Error
- Retry Required

Command Effects:

- None



Table 7-39. Send LD CXL.io Configuration Request Payload

| Byte Offset | Length in Bytes | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-------------|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0h          | 1               | <b>PPB ID:</b> Target PPB's physical port.                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 1h          | 3               | <ul style="list-style-type: none"> <li>Bits[7:0]: <b>Register Number:</b> As defined in PCIe Base Specification</li> <li>Bits[11:8]: <b>Extended Register Number:</b> As defined in PCIe Base Specification</li> <li>Bits[15:12]: <b>First Dword Byte Enable:</b> As defined in PCIe Base Specification</li> <li>Bits[22:16]: <b>Reserved</b></li> <li>Bit[23]: <b>Transaction Type:</b> <ul style="list-style-type: none"> <li>0 = Read</li> <li>1 = Write</li> </ul> </li> </ul> |
| 4h          | 2               | <b>LD ID:</b> Target LD-ID.                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 6h          | 2               | <b>Reserved</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 8h          | 4               | <b>Transaction Data:</b> Write data. Valid only for write transactions.                                                                                                                                                                                                                                                                                                                                                                                                            |

Table 7-40. Send LD CXL.io Configuration Response Payload

| Byte Offset | Length in Bytes | Description                                                      |
|-------------|-----------------|------------------------------------------------------------------|
| 0h          | 4               | <b>Return Data:</b> Read data. Valid only for read transactions. |

#### 7.6.7.3.3 Send LD CXL.io Memory Request (Opcode 5302h)

This command allows the FM to batch read or write the CXL.io Memory Space of an unbound LD or FMLD. The switch will convert the request into MemRd/MemWr TLPs to the target device. Invalid Input Return Code shall be generated if the requested LD is bound.

Possible Command Return Codes:

- Success
- Unsupported
- Invalid Input
- Internal Error
- Retry Required

Command Effects:

- None

Table 7-41. Send LD CXL.io Memory Request Payload

| Byte Offset | Length in Bytes | Description                                                                                                                                                                                                                                                                                                                                                                                                                          |
|-------------|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 00h         | 1               | <b>Port ID:</b> Target MLD port.                                                                                                                                                                                                                                                                                                                                                                                                     |
| 01h         | 2               | <ul style="list-style-type: none"> <li>Bits[11:0]: <b>Reserved</b></li> <li>Bits[15:12]: <b>First Dword Byte Enable:</b> As defined in PCIe Base Specification</li> <li>Bits[19:16]: <b>Last Dword Byte Enable:</b> As defined in PCIe Base Specification</li> <li>Bits[22:20]: <b>Reserved</b></li> <li>Bit[23]: <b>Transaction Type:</b> <ul style="list-style-type: none"> <li>0 = Read</li> <li>1 = Write</li> </ul> </li> </ul> |
| 04h         | 2               | <b>LD ID:</b> Target LD-ID.                                                                                                                                                                                                                                                                                                                                                                                                          |
| 06h         | 2               | <b>Transaction Length:</b> Transaction length in bytes, up to a maximum of 4 KB (1000h).                                                                                                                                                                                                                                                                                                                                             |
| 08h         | 8               | <b>Transaction Address:</b> The target HPA that points into the target device's MMIO Space.                                                                                                                                                                                                                                                                                                                                          |
| 10h         | Varies          | <b>Transaction Data:</b> Write data. Valid only for write transactions.                                                                                                                                                                                                                                                                                                                                                              |

Table 7-42. Send LD CXL.io Memory Request Response Payload

| Byte Offset | Length in Bytes | Description                                                      |
|-------------|-----------------|------------------------------------------------------------------|
| 0h          | 2               | <b>Return Size:</b> Number of successfully transferred bytes.    |
| 2h          | 2               | <b>Reserved</b>                                                  |
| 4h          | Varies          | <b>Return Data:</b> Read data. Valid only for read transactions. |

#### 7.6.7.4 MLD Component Command Set

This command set is only supported by, and must be supported by, MLD components implementing FM API support. These commands are processed by MLDs. When an FM is connected to a CXL switch that supports the FM API and does not have a direct connection to an MLD, these commands are passed to the MLD using the **Tunnel Management Command**. The following commands are defined:

Table 7-43. MLD Component Command Set Requirements

| Command Name         | Requirement <sup>1</sup> |
|----------------------|--------------------------|
| Get LD Info          | M                        |
| Get LD Allocations   | M                        |
| Set LD Allocations   | O                        |
| Get QoS Control      | M                        |
| Set QoS Control      | M                        |
| Get QoS Status       | O                        |
| Get QoS Allocated BW | M                        |
| Set QoS Allocated BW | M                        |
| Get QoS BW Limit     | M                        |
| Set QoS BW Limit     | M                        |

1. M = Mandatory, O = Optional.

#### 7.6.7.4.1 Get LD Info (Opcode 5400h)

This command retrieves the configurations of the MLD.

Possible Command Return Codes:

- Success
- Unsupported
- Invalid Input
- Internal Error
- Retry Required

Command Effects:

- None

**Table 7-44. Get LD Info Response Payload**

| Byte Offset | Length in Bytes | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-------------|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0h          | 8               | <b>Memory Size:</b> Total device memory capacity.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 8h          | 2               | <b>LD Count:</b> Number of logical devices supported.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Ah          | 1               | <b>QoS Telemetry Capability:</b> Optional QoS Telemetry for memory MLD capabilities for management by an FM (see <a href="#">Section 3.3.4</a> ). <ul style="list-style-type: none"> <li>• Bit[0]: <b>Egress Port Congestion Supported:</b> When set, the associated feature is supported and the Get QoS Status command must be implemented (see <a href="#">Section 3.3.4.3.9</a>).</li> <li>• Bit[1]: <b>Temporary Throughput Reduction Supported:</b> When set, the associated feature is supported (see <a href="#">Section 3.3.4.3.5</a>).</li> <li>• Bits[7:2]: <b>Reserved.</b></li> </ul> |

#### 7.6.7.4.2 Get LD Allocations (Opcode 5401h)

This command retrieves the memory allocations of the MLD.

Possible Command Return Codes:

- Success
- Unsupported
- Internal Error
- Retry Required
- Invalid Payload Length

Command Effects:

- None

**Table 7-45. Get LD Allocations Request Payload**

| Byte Offset | Length in Bytes | Description                                                                                                                    |
|-------------|-----------------|--------------------------------------------------------------------------------------------------------------------------------|
| 0h          | 1               | <b>Start LD ID:</b> ID of the first LD in the LD Allocation List.                                                              |
| 1h          | 1               | <b>LD Allocation List Limit:</b> Maximum number of LD information blocks returned. This field shall have a minimum value of 1. |

Table 7-46. Get LD Allocations Response Payload

| Byte Offset | Length in Bytes | Description                                                                                                                                                                                                                                                         |
|-------------|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0h          | 1               | <b>Number of LDs:</b> Number of LDs enabled in the device.                                                                                                                                                                                                          |
| 1h          | 1               | <b>Memory Granularity:</b> This field specifies the granularity of the memory sizes configured for each LD: <ul style="list-style-type: none"> <li>• 0h = 256 MB</li> <li>• 1h = 512 MB</li> <li>• 2h = 1 GB</li> <li>• All other encodings are reserved</li> </ul> |
| 2h          | 1               | <b>Start LD ID:</b> ID of the first LD in the LD Allocation List.                                                                                                                                                                                                   |
| 3h          | 1               | <b>LD Allocation List Length:</b> Number of LD information blocks returned. This value is the lesser of the request's 'LD Allocation List Limit' and response's 'Number of LDs'.                                                                                    |
| 4h          | Varies          | <b>LD Allocation List:</b> LD Allocation blocks for each LD, as defined in Table 7-47, repeated <b>LD Allocation List Length</b> times.                                                                                                                             |

Table 7-47. LD Allocations List Format

| Byte Offset | Length in Bytes | Description                                                                                                                                                                        |
|-------------|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0h          | 8               | <b>Range 1 Allocation Multiplier:</b> Memory Allocation Range 1 for LD. This value is multiplied with <b>Memory Granularity</b> to calculate the memory allocation range in bytes. |
| 8h          | 8               | <b>Range 2 Allocation Multiplier:</b> Memory Allocation Range 2 for LD. This value is multiplied with <b>Memory Granularity</b> to calculate the memory allocation range in bytes. |

#### 7.6.7.4.3 Set LD Allocations (Opcode 5402h)

This command sets the memory allocation for each LD. This command will fail if the device fails to allocate any of the allocations defined in the request. The allocations provided in the response reflect the state of the LD allocations after the command is processed, which allows the FM to check for partial success.

Possible Command Return Codes:

- Success
- Unsupported
- Invalid Input
- Internal Error
- Retry Required
- Invalid Payload Length

Command Effects:

- Configuration Change after Cold Reset
- Configuration Change after Conventional Reset
- Configuration Change after CXL Reset
- Immediate Configuration Change
- Immediate Data Change

Table 7-48. Set LD Allocations Request Payload

| Byte Offset | Length in Bytes | Description                                                                                                                                          |
|-------------|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0h          | 1               | <b>Number of LDs:</b> Number of LDs to configure. This field shall have a minimum value of 1.                                                        |
| 1h          | 1               | <b>Start LD ID:</b> ID of the first LD in the LD Allocation List.                                                                                    |
| 2h          | 2               | <b>Reserved</b>                                                                                                                                      |
| 4h          | Varies          | <b>LD Allocation List:</b> LD Allocation blocks for each LD, starting at Start LD ID, as defined in Table 7-47, repeated <b>Number of LDs</b> times. |

Table 7-49. Set LD Allocations Response Payload

| Byte Offset | Length in Bytes | Description                                                                                                                                                  |
|-------------|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0h          | 1               | <b>Number of LDs:</b> Number of LDs configured.                                                                                                              |
| 1h          | 1               | <b>Start LD ID:</b> ID of the first LD in the LD Allocation List.                                                                                            |
| 2h          | 2               | <b>Reserved</b>                                                                                                                                              |
| 4h          | Varies          | <b>LD Allocation List:</b> Updated LD Allocation blocks for each LD, starting at Start LD ID, as defined in Table 7-47, repeated <b>Number of LDs</b> times. |

#### 7.6.7.4.4 Get QoS Control (Opcode 5403h)

This command retrieves the MLD's QoS control parameters.

Possible Command Return Codes:

- Success
- Internal Error
- Retry Required
- Invalid Payload Length

Command Effects:

- None

Table 7-50. Payload for Get QoS Control Response, Set QoS Control Request, and Set QoS Control Response (Sheet 1 of 2)

| Byte Offset | Length in Bytes | Description                                                                                                                                                                                                                                                                                        |
|-------------|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0h          | 1               | <b>QoS Telemetry Control:</b> Default is 00h. <ul style="list-style-type: none"> <li>• Bit[0]: <b>Egress Port Congestion Enable:</b> See Section 3.3.4.3.9</li> <li>• Bit[1]: <b>Temporary Throughput Reduction Enable:</b> See Section 3.3.4.3.5</li> <li>• Bits[7:2]: <b>Reserved</b></li> </ul> |
| 1h          | 1               | <b>Egress Moderate Percentage:</b> Threshold in percent for Egress Port Congestion mechanism to indicate moderate congestion. Valid range is 1-100. Default is 10.                                                                                                                                 |
| 2h          | 1               | <b>Egress Severe Percentage:</b> Threshold in percent for Egress Port Congestion mechanism to indicate severe congestion. Valid range is 1-100. Default is 25.                                                                                                                                     |

**Table 7-50. Payload for Get QoS Control Response, Set QoS Control Request, and Set QoS Control Response (Sheet 2 of 2)**

| Byte Offset | Length in Bytes | Description                                                                                                                                                                                                                                                                        |
|-------------|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 3h          | 1               | <b>Backpressure Sample Interval:</b> Interval in ns for Egress Port Congestion mechanism to take samples. Valid range is 0-15. Default is 8 (800 ns of history for 100 samples). Value of 0 disables the mechanism. See <a href="#">Section 3.3.4.3.4</a> .                        |
| 4h          | 2               | <b>ReqCmpBasis:</b> Estimated maximum sustained sum of requests and recent responses across the entire device, serving as the basis for QoS Limit Fraction. Valid range is 0-65,535. Value of 0 disables the mechanism. Default is 0. See <a href="#">Section 3.3.4.3.7</a> .      |
| 6h          | 1               | <b>Completion Collection Interval:</b> Interval in ns for Completion Counting mechanism to collect the number of transmitted responses in a single counter. Valid range is 0-255. Default is 64 (1.024 us of history, given 16 counters). See <a href="#">Section 3.3.4.3.10</a> . |

**7.6.7.4.5 Set QoS Control (Opcode 5404h)**

This command sets the MLD's QoS control parameters, as defined in [Table 7-50](#). The device must complete the set operation before returning the response. The command response returns the resulting QoS control parameters, as defined in the same table. This command will fail, returning Invalid Input, if any of the parameters are outside their valid range.

Possible Command Codes:

- Success
- Invalid Input
- Internal Error
- Retry Required
- Invalid Payload Length

Command Effects:

- Immediate Policy Change

Payload for Set QoS Control Request and Response is documented in [Table 7-50](#).

**7.6.7.4.6 Get QoS Status (Opcode 5405h)**

This command retrieves the MLD's QoS Status. This command is mandatory if the Egress Port Congestion Supported bit is set (see [Table 7-44](#)).

Possible Command Return Codes:

- Success
- Unsupported
- Internal Error
- Retry Required
- Invalid Payload Length

Command Effects:

- None



Table 7-51. Get QoS Status Response Payload

| Byte Offset | Length in Bytes | Description                                                                                                                                      |
|-------------|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| 0h          | 1               | <b>Backpressure Average Percentage:</b> Current snapshot of the measured Egress Port average congestion. See <a href="#">Section 3.3.4.3.4</a> . |

## 7.6.7.4.7 Get QoS Allocated BW (Opcode 5406h)

This command retrieves the MLD's QoS allocated bandwidth on a per-LD basis (see [Section 3.3.4.3.7](#)).

Possible Command Return Codes:

- Success
- Invalid Input
- Internal Error
- Retry Required
- Invalid Payload Length

Command Effects:

- None

Table 7-52. Payload for Get QoS Allocated BW Request

| Byte Offset | Length in Bytes | Description                                                                              |
|-------------|-----------------|------------------------------------------------------------------------------------------|
| 0h          | 1               | <b>Number of LDs:</b> Number of LDs queried. This field shall have a minimum value of 1. |
| 1h          | 1               | <b>Start LD ID:</b> ID of the first LD in the QoS Allocated BW List.                     |

Table 7-53. Payload for Get QoS Allocated BW Response

| Byte Offset | Length in Bytes | Description                                                                                                                                                                                                                             |
|-------------|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0h          | 1               | <b>Number of LDs:</b> Number of LDs queried.                                                                                                                                                                                            |
| 1h          | 1               | <b>Start LD ID:</b> ID of the first LD in the QoS Allocated BW List.                                                                                                                                                                    |
| 2h          | Number of LDs   | <b>QoS Allocation Fraction:</b> Byte array of allocated bandwidth fractions for LDs, starting at Start LD ID. The valid range of each array element is 0-255. Default value is 0. Value in each byte is the fraction multiplied by 256. |

## 7.6.7.4.8 Set QoS Allocated BW (Opcode 5407h)

This command sets the MLD's QoS allocated bandwidth on a per-LD basis, as defined in [Section 3.3.4.3.7](#). The device must complete the set operation before returning the response. The command response returns the resulting QoS allocated bandwidth, as defined in the same table. This command will fail, returning Invalid Input, if any of the parameters are outside their valid range.

Possible Command Return Codes:

- Success
- Invalid Input
- Internal Error

- Retry Required
- Invalid Payload Length

Command Effects:

- Configuration Change after Cold Reset
- Configuration Change after Conventional Reset
- Configuration Change after CXL Reset
- Immediate Configuration Change
- Immediate Data Change

**Table 7-54. Payload for Set QoS Allocated BW Request, and Set QoS Allocated BW Response**

| Byte Offset | Length in Bytes | Description                                                                                                                                                                                                                             |
|-------------|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0h          | 1               | <b>Number of LDs:</b> Number of LDs configured.                                                                                                                                                                                         |
| 1h          | 1               | <b>Start LD ID:</b> ID of the first LD in the QoS Allocated BW List.                                                                                                                                                                    |
| 2h          | Number of LDs   | <b>QoS Allocation Fraction:</b> Byte array of allocated bandwidth fractions for LDs, starting at Start LD ID. The valid range of each array element is 0-255. Default value is 0. Value in each byte is the fraction multiplied by 256. |

#### 7.6.7.4.9 Get QoS BW Limit (Opcode 5408h)

This command retrieves the MLD's QoS bandwidth limit on a per-LD basis (see [Section 3.3.4.3.7](#)).

Possible Command Return Codes:

- Success
- Invalid Input
- Internal Error
- Retry Required
- Invalid Payload Length

Command Effects:

- None

**Table 7-55. Payload for Get QoS BW Limit Request**

| Byte Offset | Length in Bytes | Description                                                                              |
|-------------|-----------------|------------------------------------------------------------------------------------------|
| 0h          | 1               | <b>Number of LDs:</b> Number of LDs queried. This field shall have a minimum value of 1. |
| 1h          | 1               | <b>Start LD ID:</b> ID of the first LD in the QoS BW Limit List.                         |

**Table 7-56. Payload for Get QoS BW Limit Response**

| Byte Offset | Length in Bytes | Description                                                                                                                                                                                                                              |
|-------------|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0h          | 1               | <b>Number of LDs:</b> Number of LDs queried.                                                                                                                                                                                             |
| 1h          | 1               | <b>Start LD ID:</b> ID of the first LD in the QoS BW Limit List.                                                                                                                                                                         |
| 2h          | Number of LDs   | <b>QoS Limit Fraction:</b> Byte array of allocated bandwidth limit fractions for LDs, starting at Start LD ID. The valid range of each array element is 0-255. Default value is 0. Value in each byte is the fraction multiplied by 256. |

#### 7.6.7.4.10 Set QoS BW Limit (Opcode 5409h)

This command sets the MLD's QoS bandwidth limit on a per-LD basis, as defined in [Section 3.3.4.3.7](#). The device must complete the set operation before returning the response. The command response returns the resulting QoS bandwidth limit, as defined in the same table. This command will fail, returning Invalid Input, if any of the parameters are outside their valid range. This command will fail, returning Internal Error, if the device was able to set the QoS BW Limit for some of the LDs in the request, but not all the LDs.

Possible Command Return Codes:

- Success
- Invalid Input
- Internal Error
- Retry Required
- Invalid Payload Length

Command Effects:

- Configuration Change after Cold Reset
- Configuration Change after Conventional Reset
- Configuration Change after CXL Reset
- Immediate Configuration Change
- Immediate Data Change

**Table 7-57. Payload for Set QoS BW Limit Request, and Set QoS BW Limit Response**

| Byte Offset | Length in Bytes | Description                                                                                                                                                                                                                              |
|-------------|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0h          | 1               | <b>Number of LDs:</b> Number of LDs configured.                                                                                                                                                                                          |
| 1h          | 1               | <b>Start LD ID:</b> ID of the first LD in the QoS BW Limit List.                                                                                                                                                                         |
| 2h          | Number of LDs   | <b>QoS Limit Fraction:</b> Byte array of allocated bandwidth limit fractions for LDs, starting at Start LD ID. The valid range of each array element is 0-255. Default value is 0. Value in each byte is the fraction multiplied by 256. |

#### 7.6.7.5 Multi-Headed Device Command Set

The Multi-Headed device command set includes commands for querying the Head-to-LD mapping in a Multi-Headed device. Support for this command set is required on the LD Pool CCI of a Multi-Headed device.

##### 7.6.7.5.1 Get Multi-Headed Info (Opcode 5500h)

This command retrieves the number of heads, number of supported LDs, and Head-to-LD mapping of a Multi-Headed device.

Possible Command Return Codes:

- Success
- Unsupported
- Invalid Input
- Internal Error
- Retry Required

Command Effects:

- None

**Table 7-58. Get Multi-Headed Info Request Payload**

| Byte Offset | Length in Bytes | Description                                                                                                      |
|-------------|-----------------|------------------------------------------------------------------------------------------------------------------|
| 0h          | 1               | <b>Start LD ID:</b> ID of the first LD in the LD Map.                                                            |
| 1h          | 1               | <b>LD Map List Limit:</b> Maximum number of LD Map entries returned. This field shall have a minimum value of 1. |

**Table 7-59. Get Multi-Headed Info Response Payload**

| Byte Offset | Length in Bytes | Description                                                                                                                                                                                       |
|-------------|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0h          | 1               | <b>Number of LDs:</b> Total number of LDs in the LD Pool. This field shall have a minimum value of 1.                                                                                             |
| 1h          | 1               | <b>Number of Heads:</b> Total number of CXL heads. This field shall have a minimum value of 1.                                                                                                    |
| 2h          | 2               | <b>Reserved</b>                                                                                                                                                                                   |
| 4h          | 1               | <b>Start LD ID:</b> ID of the first LD in the LD Map.                                                                                                                                             |
| 5h          | 1               | <b>LD Map Length:</b> Number of LD Map entries returned.<br>LD Map Length = Min (LD Map List Limit, (Number of LDs - Start LD ID))                                                                |
| 6h          | 2               | <b>Reserved</b>                                                                                                                                                                                   |
| 8h          | LD Map Length   | <b>LD Map:</b> Port number of the head to which each LD is assigned, starting at Start LD ID, repeated LD Map Length times. A value of FFh indicates that LD is not currently assigned to a head. |

#### 7.6.7.5.2 Get Head Info (Opcode 5501h)

This command retrieves information for one or more heads.

This command fails with the Invalid Input return code if the values of the Start Head and Number of Heads fields request the information for a non-existent head.

Possible Command Return Codes:

- Success
- Unsupported
- Invalid Input
- Internal Error
- Retry Required

Command Effects:

- None

**Table 7-60. Get Head Info Request Payload**

| Byte Offset | Length in Bytes | Description                                                                        |
|-------------|-----------------|------------------------------------------------------------------------------------|
| 0h          | 1               | <b>Start Head:</b> Specifies the ID of the first head information block requested. |
| 1h          | 1               | <b>Number of Heads:</b> Number of head information blocks requested.               |

Table 7-61. Get Head Info Response Payload

| Byte Offset | Length in Bytes | Description                                                                                                           |
|-------------|-----------------|-----------------------------------------------------------------------------------------------------------------------|
| 0h          | 1               | <b>Number of Heads:</b> Number of head information blocks returned.                                                   |
| 1h          | 3               | <b>Reserved</b>                                                                                                       |
| 4h          | Varies          | <b>Head Information List:</b> Head information block as defined in Table 7-62, repeated <b>Number of Heads</b> times. |

Table 7-62. Get Head Info Head Information Block Format

| Byte Offset | Length in Bytes | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|-------------|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0h          | 1               | <b>Port Number:</b> Value encoding matches the Port Number field in the PCIe Link Capabilities register in the PCIe Capability structure.                                                                                                                                                                                                                                                                                                                                               |
| 1h          | 1               | <ul style="list-style-type: none"> <li>Bits[5:0]: <b>Maximum Link Width:</b> Value encoding matches the Maximum Link Width field in the PCIe Link Capabilities register in the PCIe Capability structure</li> <li>Bits[7:6]: <b>Reserved</b></li> </ul>                                                                                                                                                                                                                                 |
| 2h          | 1               | <ul style="list-style-type: none"> <li>Bits[5:0]: <b>Negotiated Link Width:</b> Value encoding matches the Negotiated Link Width field in the PCIe Link Capabilities register in the PCIe Capability structure</li> <li>Bits[7:6]: <b>Reserved</b></li> </ul>                                                                                                                                                                                                                           |
| 3h          | 1               | <ul style="list-style-type: none"> <li>Bits[5:0]: <b>Supported Link Speeds Vector:</b> Value encoding matches the Supported Link Speeds Vector field in the PCIe Link Capabilities 2 register in the PCIe Capability structure</li> <li>Bits[7:6]: <b>Reserved</b></li> </ul>                                                                                                                                                                                                           |
| 4h          | 1               | <ul style="list-style-type: none"> <li>Bits[5:0]: <b>Max Link Speed:</b> Value encoding matches the Max Link Speed field in the PCIe Link Capabilities register in the PCIe Capability structure</li> <li>Bits[7:6]: <b>Reserved</b></li> </ul>                                                                                                                                                                                                                                         |
| 5h          | 1               | <ul style="list-style-type: none"> <li>Bits[5:0]: <b>Current Link Speed:</b> Value encoding matches the Current Link Speed field in the PCIe Link Status register in the PCIe Capability structure</li> <li>Bits[7:6]: <b>Reserved</b></li> </ul>                                                                                                                                                                                                                                       |
| 6h          | 1               | <p><b>LTSSM State:</b> Current link LTSSM Major state:</p> <ul style="list-style-type: none"> <li>00h = Detect</li> <li>01h = Polling</li> <li>02h = Configuration</li> <li>03h = Recovery</li> <li>04h = L0</li> <li>05h = L0s</li> <li>06h = L1</li> <li>07h = L2</li> <li>08h = Disabled</li> <li>09h = Loopback</li> <li>0Ah = Hot Reset</li> <li>All other encodings are reserved</li> </ul> <p>Link substates should be reported through vendor-defined diagnostics commands.</p> |
| 7h          | 1               | <b>First Negotiated Lane Number</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 8h          | 1               | <p><b>Link State Flags</b></p> <ul style="list-style-type: none"> <li>Bit[0]: <b>Lane Reversal State:</b> <ul style="list-style-type: none"> <li>0 = Standard lane ordering</li> <li>1 = Reversed lane ordering</li> </ul> </li> <li>Bit[1]: <b>Port PCIe Reset State (PERST#):</b> <ul style="list-style-type: none"> <li>0 = Not in reset</li> <li>1 = In reset</li> </ul> </li> <li>Bits[7:2]: <b>Reserved</b></li> </ul>                                                            |

### 7.6.7.6 DCD Management Command Set for LD-FAM

The DCD Management command set, described in the following subsections, includes commands for querying and configuring Dynamic Capacity for LD-FAM (SLDs and MLDs). It is used by the FM to manage memory assignment within an LD-FAM DCD. Memory management for G-FAM (GFDs) is defined in [Section 8.2.10.9.10](#).

#### 7.6.7.6.1 Get DCD Info (Opcode 5600h)

This command retrieves the number of supported hosts, total Dynamic Capacity of the device, and supported region configurations for an LD-FAM DCD. To retrieve the corresponding DCD info for a GFD, see [Section 8.2.10.9.10.1](#).

Possible Command Return Codes:

- Success
- Unsupported
- Internal Error
- Retry Required

Command Effects:

- None

**Table 7-63. Get DCD Info Response Payload (Sheet 1 of 2)**

| Byte Offset | Length in Bytes | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|-------------|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 00h         | 1               | <b>Number of Hosts:</b> Total number of hosts that the device supports. This field shall have a minimum value of 1.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 01h         | 1               | <b>Number of Supported DC Regions:</b> The device shall report the total number of Dynamic Capacity Regions available per LD. DCDs shall report between 1 and 8 regions. All other encodings are reserved.                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 02h         | 2               | <b>Reserved</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 04h         | 2               | <ul style="list-style-type: none"> <li>• <b>Bits[3:0]: Supported Add Capacity Selection Policies:</b> Bitmask that specifies the selection policies, as defined in <a href="#">Section 7.6.7.6.5</a>, that the device supports when capacity is added. At least one policy shall be supported. A value of 1 indicates that a policy is supported, and a value of 0 indicates that a policy is not supported: <ul style="list-style-type: none"> <li>— Bit[0]: <b>Free</b></li> <li>— Bit[1]: <b>Contiguous</b></li> <li>— Bit[2]: <b>Prescriptive</b></li> <li>— Bit[3]: Must be 0</li> </ul> </li> <li>• <b>Bits[15:4]: Reserved</b></li> </ul> |
| 06h         | 2               | <b>Reserved</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 08h         | 2               | <ul style="list-style-type: none"> <li>• <b>Bits[1:0]: Supported Release Capacity Removal Policies:</b> Bitmask that specifies the removal policies, as defined in <a href="#">Section 7.6.7.6.6</a>, that the device supports when capacity is released. At least one policy shall be supported. A value of 1 indicates that a policy is supported, and a value of 0 indicates that a policy is not supported: <ul style="list-style-type: none"> <li>— Bit[0]: <b>Tag-based</b></li> <li>— Bit[1]: <b>Prescriptive</b></li> </ul> </li> <li>• <b>Bits[15:2]: Reserved</b></li> </ul>                                                           |
| 0Ah         | 1               | <b>Sanitize on Release Configuration Support Mask:</b> Bitmask, where bit position corresponds to region number, indicating whether the Sanitize on Release capability is configurable (1) or not configurable (0) for that region.                                                                                                                                                                                                                                                                                                                                                                                                              |
| 0Bh         | 1               | <b>Reserved</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 0Ch         | 8               | <b>Total Dynamic Capacity:</b> Total memory media capacity of the device available for dynamic assignment to any host in multiples of 256 MB.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |